Sangjoon "Joon" Lee, Ph.D.

Assistant Professor at Nanyang Technological University (NTU Singapore)

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Assistant professor of **mechanical and aerospace engineering**, with a core focus on **high-fidelity CFD simulations coupled with data-driven surrogate models for physics-informed, cost-efficient aerodynamic prediction**. Expertise in leveraging high-performance computing and machine-learning inference techniques to pioneer **optimized engineering designs in aerospace systems**. Committed to advancing the understanding of **fluid and aeroscience across scales**, ranging from heat exchangers, pipe flows, and HVAC systems to turbines, aircraft, and environmental flows.

WORK EXPERIENCE

Nanyang Technological University – Singapore — 2026/11 –
Assistant Professor, School of Mechanical and Aerospace Engineering
[2026/11 – Present]

Stanford University – Stanford, CA, United States — 2024/10 – 2026/10
Postdoctoral Scholar, Aerospace Design Lab
[2026/01 – 2026/10]
Postdoctoral Fellow, Center for Turbulence Research (CTR)
[2024/10 – 2025/12]

University of California, Berkeley – Berkeley, CA, United States — 2020/01 – 2024/08
Graduate Student Researcher, Computational Fluid Dynamics (CFD) Lab
[2020/01 – 2024/08]

Seoul National University – Seoul, South Korea — 2016/09 – 2018/12
Researcher, Energy & Environmental Flow Lab (EEFL)
[2017/09 – 2018/12]
Research Intern, Turbulence, Flow Control & CFD Lab
[2016/09 – 2017/12]

EDUCATION

University of California, Berkeley – Berkeley, CA, United States — 2019/08 – 2024/08
Ph.D. / M.S. in Mechanical Engineering
- Designated emphasis: Computational and Data Science and Engineering

Seoul National University – Seoul, South Korea — 2012/03 – 2018/08
(involving 21-month military leave)
B.S. in Mechanical and Aerospace Engineering &
B.B.A. (Bachelor of Business Administration)
- Honors: *Summa Cum Laude*

Seoul Science High School – Seoul, South Korea — 2009/03 – 2012/02

RESEARCH GRANTS

Precourt Institute for Energy Seed Grant: SMC-001 — 2026/02 – 2027/01
Sustainable Mobility Center at Stanford University
Co-PI · $100,000
- Highly efficient thermal management in hydrogen-powered aviation

NSF ACCESS Allocation: PHY250071 2025/03 – 2027/02
ACCESS by the U.S. National Science Foundation (NSF)
PI · 1,500,000 HPC core-hours
- Surface topography optimization for thermally-efficient dropwise condensation

FELLOWSHIPS

CTR Postdoctoral Fellowship ($100,000) 2024/10 – 2025/12
Center for Turbulence Research (CTR) at Stanford University

Departmental Graduate Fellowship ($23,825) 2023/08 – 2023/12
College of Engineering at the University of California, Berkeley

Ilju Overseas Ph.D. Scholarship ($120,000), Study Abroad Doctoral Program 2019/08 – 2023/07
Ilju Academy & Culture Foundation
- Merit-based; selected as one of six recipients in 2019

National Scholarship for Science and Engineering (Full-tuition) 2012/03 – 2017/12
Korea Student Aid Foundation (KOSAF)

HONORS & AWARDS

Outstanding Graduate Student Instructor (OGSI) Award 2021
GSI Teaching & Resource Center at the University of California, Berkeley

Representative of the Engineering Class of 2018, 72nd Summer Commencement 2018
Seoul National University

TEACHING & TUTORING

University of California, Berkeley
Graduate Student Instructor (GSI)
- Computer Programming for Scientists and Engineers (ENGIN 7) 2024 Sp
- Aerospace Engineering Design (AERO ENG 10) 2022 Fa – 2023 Sp
- Experimentation and Measurements (MEC ENG 103) 2019 Fa – 2022 Sp

Seoul National University
Undergraduate Tutor
- Basic Calculus 1, 2 (007.098A, 007.102) 2013 Sp – 2013 Fa
- Basic Physics 1 (007.099A) 2013 Sp

PROFESSIONAL SERVICE

Peer Reviewer
- Transactions on Aerospace Research (Łukasiewicz – ILOT) 2026 –
- Journal of Building Performance Simulation (Taylor & Francis) 2026 –
- Journal of Computational Physics (Elsevier) 2026 –
- Journal of Fluid Mechanics (Cambridge University Press) 2025 –
- Physics of Fluids (AIP Publishing) 2024 –

COMMUNITY OUTREACH

Stanford seeME & CTR^2, Stanford University 2025
Volunteer (Teaching, Photographing & On-day Assistance)
- On-campus hands-on classes for young students to learn various aspects of engineering

SNU Tomorrow's Engineers Membership, Seoul National University 2016 – 2018
Member & Head Manager
- Annual *Vision Mentoring* events for high school students interested in engineering and science

ACADEMIC PUBLICATIONS

PEER-REVIEWED ARTICLES

* corresponding
† co-first

1. Vijay, S., & Lee, S.* (2027). **Topology-Aware Modeling of Isotropic Permeability in Structured Porous Media**. In Preparation.
2. Lee, S., & Sheikh, H. M. (2027). **Riblet Multi-Objective Optimization**. In Preparation.
3. Jung, J., & Lee, S.* (2027). **Data-Driven Optimization of Novel Morphing Airfoil Designs for Enhanced Flutter Control**. In Preparation.
4. Hong, J., Lee, S.†, Lee, D., Bae, J., & Hwang, W. (2027). **Investigation of Turbulent Flow over Riblet Surfaces using Magnetic Resonance Velocimetry and Large Eddy Simulation**. In Preparation.
5. Lee, S.*, & Wang, J. (2027). **MLegS: A Modern Mapped Legendre Spectral Method Solver for Unbounded Domains with Parallelization**. In Preparation. arXiv:2609.18026.
6. Lim, D. D., Yoon, J., Choi, H., Soh, J., Chung, R., Kim, W., Choi, B., Kim, W., Park, J., Lee, S., Gu, G. X., & Min, J.-S. (2027). **Prehensile-Tail-Inspired Normally Locked Joints for Selective and Power-Free Shape Reconfiguration**. *Nature Communications*, Under Review.
7. Lee, S.*, & Sheikh, H. M. (2027). **A Physics-in-the-Loop Benchmark for Single- and Multi-Objective Optimization: An Airfoil Design Case and Dataset**. *PeerJ Computer Science*, Under Review.
8. Cheng, Z., Dowell, E. H., Lee, S., & Gu, G. X. (2027). **Internal Topological Design Regulates Elastic-Energy Localization for Passive Aeroelastic Control**. *Proceedings of the National Academy of Sciences*, Under Review.
9. Lee, S., Yildizdag, M. E., & Sheikh, H. M. (2026). **Bayesian Optimization and Topographic Exploration of Drag-Reducing Dimples for Aerodynamic Surfaces**. *Aerospace Science and Technology*, Under Review. arXiv:2608.12826.
10. Wang, J., Lee, S., & Marcus, P. S. (2026). **Perturbation Analysis of Triadic Resonance in Columnar Vortices: Selection Rules and the Roles of External Forcing and Critical Layers**. *Journal of Fluid Mechanics*, Accepted. arXiv:2402.05287.
11. Duarte, C., Raftery, P., Lee, S., & Solmaz, A. S. (2026). **Room Surface Convective Heat Transfer with Ceiling Fans and Its Effect on Radiant Cooling Systems**. *Journal of Building Performance Simulation*. https://doi.org/10.1080/19401493.2026.2656928.
12. Lee, S., & Sheikh, H. M. (2026). **Airfoil Optimization using Design-by-Morphing with Minimized Design-Space Dimensionality**. *Journal of Computational Design and Engineering*, 13(1), 108–124. https://doi.org/10.1093/jcde/qwaf124.
13. Lee, S., Song, H., & Lele, S. K. (2025). **Global Stability Analysis for Multidimensional Flow using an Augmented State Vector Formulation**. In *CTR Annual Research Briefs 2025* (pp. 313–323). Center for Turbulence Research, Stanford University. https://web.stanford.edu/group/ctr/ResBriefs/2025/28_Lee.
14. Lee, S., & Vijay, S. (2025). **Topology-Aware Permeability Modeling in Structured Porous Media for Passive Flow Control**. In *CTR Annual Research Briefs 2025* (pp. 351–361). Center for Turbulence Research, Stanford University. https://web.stanford.edu/group/ctr/ResBriefs/2025/31_Lee.

15. Lee, S., & Marcus, P. S. (2025). **Transient Growth of a Wake Vortex and its Initiation via Inertial Particles**. *Journal of Fluid Mechanics*, 1014, A16. https://doi.org/10.1017/jfm.2025.253.

16. Lee, S., Baek, S., Ryu, J., Song, M., & Hwang, W. (2025). **Flow in Ribbed Cooling Channels with Additive Manufacturing-Induced Surface Roughness**. *Physics of Fluids*, 37(6), 065118. https://doi.org/10.1063/5.0268180.

17. Lee, S.*, Sheikh, H. M., Lim, D. D., Gu, G. X., & Marcus, P. S. (2024). **Bayesian-Optimized Riblet Surface Design for Turbulent Drag Reduction via Design-by-Morphing with Large Eddy Simulation**. *Journal of Mechanical Design*, 146(8), 081701. https://doi.org/10.1115/1.4064413.

18. Lee, S., & Marcus, P. S. (2023). **Linear Stability Analysis of Wake Vortices by a Spectral Method Using Mapped Legendre Functions**. *Journal of Fluid Mechanics*, 967, A2. https://doi.org/10.1017/jfm.2023.455.

19. Sheikh, H. M., Lee, S.†, Wang, J., & Marcus, P. S. (2023). **Airfoil Optimization using Design-by-Morphing**. *Journal of Computational Design and Engineering*, 10(4), 1443–1459. https://doi.org/10.1093/jcde/qwad059.

20. Lee, S., & Hwang, W. (2019). **Development of an Efficient Immersed-Boundary Method with Subgrid-Scale Models for Conjugate Heat Transfer Analysis using Large Eddy Simulation**. *International Journal of Heat and Mass Transfer*, 134, 198–208. https://doi.org/10.1016/j.ijheatmasstransfer.2019.01.019.

21. Baek, S., Lee, S., Hwang, W., & Park, J. S. (2019). **Experimental and Numerical Investigation of the Flow in a Trailing Edge Ribbed Internal Cooling Passage**. *Journal of Turbomachinery*, 141(1), 011012. https://doi.org/10.1115/1.4041868.

CONFERENCE PAPERS & PRESENTATIONS

1. Lee, S., & Alonso, J. J. (2027, Jan 11–15). **Topographic Design of Fuel-Cell Cooling Passages for Hydrogen Aviation**. AIAA SciTech 2027 Forum, Orlando, FL, United States. American Institute of Aeronautics and Astronautics, Accepted.

2. Lee, S., & Song, H. (2027, Jan 11–15). **GPU-Accelerated Global Stability Analysis for Multi-dimensional Fluid Flows**. AIAA SciTech 2027 Forum, Orlando, FL, United States. American Institute of Aeronautics and Astronautics, Accepted.

3. Park, J., Lee, S., Li, J., Chen, K. W., Izuhara, I., & Schiavon, S. (2026, May 18–22). **A Simulation Study on Condensation Risk in Radiant Cooling Panels with Elevated Air Movement**. 12th International Conference on Indoor Air Quality, Ventilation & Energy Conservation in Buildings, Los Angeles, CA, United States (no. 02027). IAQVEC Association. https://doi.org/10.1051/e3sconf/202671602027. *Best Paper Award*.

4. Lee, S., & Vijay, S. (2025, Nov 23–25). **Topological Design of Porous Structures for Flow Control: A Design-by-Morphing Approach**. 78th Annual Meeting of the APS Division of Fluid Dynamics (APS-DFD), Houston, TX, United States (no. J32.3). American Physical Society.

5. Hong, J., Lee, S., Lee, D., Bae, J., & Hwang, W. (2025, Sep 15–19). **Experimental and Numerical Investigation of 3D Flow Structures in a Turbulent Channel Flow with Riblets**. 16th International Symposium on Experimental and Computational Aerothermodynamics of Internal Flows (ISAIF), Prague, Czechia (no. C7.1). Institute of Thermomechanics, Czech Academy of Sciences.

6. Lee, S., Wang, J., & Marcus, P. S. (2024, Nov 24–26). **Modernized and Parallelized Mapped Legendre Spectral Method Code for Unbounded Vortical Flow Simulations**. 77th Annual Meeting of the APS Division of Fluid Dynamics (APS-DFD), Salt Lake City, UT, United States (no. L16.7). American Physical Society.

7. Wang, J., Lee, S., & Marcus, P. S. (2024, Nov 24–26). **Stability Analysis of the Q-Vortex: Critical Swirling Parameter Determination via Perturbation Theories and Resonant Triadic Perturbations in the Sub-Critical Region**. 77th Annual Meeting of the APS Division of Fluid Dynamics (APS-DFD), Salt Lake City, UT, United States (no. J38.5). American Physical Society.

8. Lee, S., & Marcus, P. S. (2024, Aug 25–30). **Particle-Initiated Transient Growth of a Wake Vortex in Consideration of Condensation Trails**. 26th International Congress of Theoretical and Applied Mechanics (ICTAM), Daegu, South Korea (pp. 2009–2010). International Union of Theoretical and Applied Mechanics. https://doi.org/10.979.11984760/98.

9. Lee, S., & Marcus, P. S. (2023, Nov 19–21). **Investigation of Triggering Vortex Instabilities with Inertial Particles**. 76th Annual Meeting of the APS Division of Fluid Dynamics (APS-DFD), Washington, DC, United States (no. ZC38.5). American Physical Society.

10. Wang, J., Lee, S., & Marcus, P. S. (2023, Nov 19–21). **Three-Wave Resonance in Neutrally Stable Wake Vortices**. 76th Annual Meeting of the APS Division of Fluid Dynamics (APS-DFD), Washington, DC, United States (no. ZC38.2). American Physical Society.

11. Lee, S., & Marcus, P. S. (2022, Nov 20–22). **Viscous Perturbation to Inviscid Wake Vortices: Perturbation Theory in Vortex Stability**. 75th Annual Meeting of the APS Division of Fluid Dynamics (APS-DFD), Indianapolis, IN, United States (no. Q11.7). American Physical Society.

12. Marcus, P. S., Wang, J., & Lee, S. (2022, Nov 20–22). **A General Framework for Destabilizing Neutrally-Stable Flows Applied to Aircraft Wake Vortices**. 75th Annual Meeting of the APS Division of Fluid Dynamics (APS-DFD), Indianapolis, IN, United States (no. L18.1). American Physical Society.

13. Lee, S., & Marcus, P. S. (2021, Nov 21–23). **Linear Instability Analysis of Wake Vortices by a Spectral Method using Mapped Legendre Functions**. 74th Annual Meeting of the APS Division of Fluid Dynamics (APS-DFD), Phoenix, AZ, United States (no. E24.1). American Physical Society.

14. Wang, J., Lee, S., & Marcus, P. S. (2021, Nov 21–23). **Destabilizing Neutrally Stable Wake Vortices Using Degenerate Eigenmodes**. 74th Annual Meeting of the APS Division of Fluid Dynamics (APS-DFD), Phoenix, AZ, United States (no. E24.3). American Physical Society.

15. Lee, S., & Hwang, W. (2018, Jul 4–6). **Validation of a Conjugate Heat Transfer Code with Subgrid-Scale Models for Turbulent Flow**. KSFM 2018 Summer Conference, Jeju, South Korea (pp. 197–198). Korean Society for Fluid Machinery.

16. Baek, S., Lee, S., Hwang, W., & Park, J. S. (2018, Jun 11–15). **Experimental and Numerical Investigation of the Flow in a Trailing Edge Ribbed Internal Cooling Passage**. ASME 2018 Turbo Expo: Turbomachinery Technical Conference and Exposition, Lillestrøm, Norway (no. GT2018-76741). American Society of Mechanical Engineers. https://doi.org/10.1115/GT2018-76741. *Journal-Quality Appraisal and Transferred to J. Turbomach.*

17. Lee, S. (2017, Nov 1–3). **2D Simulation of an Unsteady Flow around a Small Vertical Axis Wind Turbine Using an Immersed Boundary Method**. KSME 2017 Annual Conference, Jeju, South Korea (pp. 741–745). Korean Society of Mechanical Engineers. *Student Paper Award: Bronze*.

18. Baek, S., Lee, S., & Hwang, W. (2017, Nov 1–3). **Investigation of Fully Developed Turbulent Pipe Flow Using Magnetic Resonance Velocimetry (MRV) and Large Eddy Simulation (LES)**. KSME 2017 Annual Conference, Jeju, South Korea (pp. 581–583). Korean Society of Mechanical Engineers.

INVITED TALKS & SEMINARS

1. Lee, S. (2026, Jul 30). **Thermofluid Design of Fuel Cell Cooling Passages for Hydrogen Powered Aviation**. AeroAstro Summer Research Seminar Series. Department of Aeronautics and Astronautics, Stanford University, Stanford, CA, United States.
2. Lee, S. (2026, Feb 5). **Training-Free Design Generation through Morphing and Its Applications in Fluid Dynamics**. MAE 297 Seminar Series. Department of Mechanical and Aerospace Engineering, University of California, Davis, Davis, CA, United States.
3. Lee, S. (2025, Aug 19). **Topology Optimization of Complex Nonlinear Systems Using High-Performance Simulations and Data-Driven Approaches**. GTR Technical Research Society Seminar. Global Technology Research, Samsung Electronics, Suwon, South Korea.
4. Lee, S. (2025, Jan 10). **Unmasking Hidden Physics and Bridging Data Sparsity: Two Paths to Tackling Fluid Problems**. CTR Tea Seminar. Center for Turbulence Research, Stanford University, Stanford, CA, United States.
5. Lee, S. (2024, Sep 10). **Physics-Based Computation in the Modern Era of Data-Driven Fluid Mechanics**. SNU Mechanical Engineering Seminar. Department of Mechanical Engineering, Seoul National University, Seoul, South Korea.
6. Lee, S. (2023, Aug 8). **Design-by-Morphing (DbM): A Novel Design Methodology for Aerodynamic Optimization**. 2023 Hyundai Vision Conference. Hyundai Motor Company, Seoul, South Korea.
7. Lee, S. (2022, Nov 16). **Modern Applications of Computational Fluid Dynamics (CFD)**. 2022 Online Special Lecture Series: Research Reinforcement for Sustainable Buildings and Urban Systems in Future. Department of Architectural and Urban Systems Engineering, Ewha Womans University, Online.
8. Lee, S. (2018, Aug 8). **An Introduction to In-House LES: Applications to Turbine Internal Cooling and Recent Improvements for Conjugate Heat Transfer Analysis**. KARI Computational Fluid Dynamics Seminar. Korea Aerospace Research Institute, Daejeon, South Korea.

SOFTWARE & DATA

1. Lee, S., & Sheikh, H. M. (2026). **AirDbM-Bench: Python API for AirDbM with Single-/Multi-Objective Optimization Benchmarks** (v1.0). https://github.com/jun9303/AirDbM-Bench.
2. Lee, S., & Wang, J. (2026). **MLegS: Modernized and Parallelized Mapped Legendre Spectral Method Code** (v1.1.3). https://github.com/ucbCFD/MLegS.
3. Lee, S. (2025). **roughSurfaceGen: Artificial Rough Surface Generator that Fits Prescribed Surface Roughness Parameters** (v1.0.1). https://github.com/jun9303/roughSurfaceGen.